CancerOS — Clinical Data & Dataset Specification

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1. Purpose

This specification defines how CancerOS will discover, acquire, import, validate, normalize, store, link, and use clinical research data.

It establishes the authoritative data structure shared by the imaging engine, synthetic longitudinal simulator, temporal analysis engine, knowledge layer, and user interface.

The system must preserve authentic patient data while permitting synthetic observations to be generated as separate, traceable research branches.

1.1 Primary objectives

1. Establish reliable breast cancer imaging and clinical data sources.
2. Define a canonical patient research data model.
3. Standardize imaging, laboratory, pathology, treatment, and clinical observation formats.
4. Establish data lineage, licensing, and access restrictions.
5. Distinguish authentic observations from derived and simulated records.
6. Prevent source duplication and training/evaluation leakage.
7. Provide sufficiently structured inputs for subsequent synthetic longitudinal simulation.

1.2 Design principles

- **Source fidelity:** Original clinical facts must never be silently rewritten.
- **Provenance first:** Every observation must retain its source and generation method.
- **Modality separation:** An MRI acquisition, clinical visit, blood test, and treatment event are different entities.

- **Patient-level linkage:** Records must be associated through verified pseudonymous research identifiers.
- **Missingness preservation:** Missing values must not be treated as normal findings.
- **Temporal accuracy:** Actual source dates and synthetic dates must remain distinguishable.
- **Scientific reproducibility:** Data selection, transformation, and model evaluation must be repeatable.
- **Minimal exposure:** Keep source images and clinical data protected; store only data necessary for the prototype.

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2. Dataset Source Inventory

DS-01 — I-SPY2

Provider: The Cancer Imaging Archive (TCIA)
Collection: I-SPY2
URL: https://www.cancerimagingarchive.net/collection/ispy2/
Collection DOI: https://doi.org/10.7937/TCIA.D8Z0-9T26
Primary collection: 719 patients
Broader I-SPY2 Imaging Cohort 1: 985 patients, comprising 719 principal I-SPY2 cases and 266 ACRIN-6698 cases
Principal image collection size: Approximately 1.75 TB
License: Creative Commons Attribution 4.0 (CC BY 4.0)

Verified data characteristics

- Breast dynamic contrast-enhanced MRI.
- Actual repeated imaging during neoadjuvant chemotherapy.
- Pretreatment MRI (T0).
- Early-treatment MRI (T1).
- Mid-treatment MRI (T2).
- Post-treatment, presurgical MRI (T3).
- Histopathologic outcome information.
- Selected demographic, molecular, treatment, and follow-up metadata.

- Imaging-derived quantitative measurements.
- DICOM imaging and selected segmentation or analysis-mask resources.

Four time points represent the trial protocol design, not a guarantee that every patient has four complete usable imaging sessions.

CancerOS role

Primary source for genuine longitudinal tumor visualization and temporal evaluation.

Required verification

- Confirm matching patient identifiers across image studies and clinical spreadsheets.
- Confirm actual scan timing for each patient.
- Inspect available tumor annotations and derived measurements.
- Review treatment coding before mapping to CancerOS treatment categories.
- Determine which laboratory or symptom variables, if any, are actually provided.
- Verify that the selected patients have adequate imaging and registration quality.

DS-02 — I-SPY1

Provider: TCIA
Collection: I-SPY1 / ACRIN 6657
URL: https://www.cancerimagingarchive.net/collection/ispy1/
DOI: https://doi.org/10.7937/K9/TCIA.2016.HdHpgJLK
Participants: 222
MRI studies: 847
Collection size: Approximately 78 GB
License: CC BY 3.0 for the principal collection; each downloaded component must be checked individually.

Verified data characteristics

- Multiple MRI examinations.
- Breast DCE-MRI.
- Selected segmentation or image-analysis data.
- Clinical and follow-up spreadsheets.
- Tumor longest-diameter measurements for certain curated subsets.

- Outcome-related information in selected subsets.

CancerOS role

Secondary real longitudinal dataset for integration tests, external evaluation, and replication of measurements.

Required verification

- Identify patients with matched clinical records and repeated scans.
- Check usable segmentation and image-analysis annotations.
- Confirm measurement definitions and time points.
- Avoid assuming every participant has all observations.

DS-03 — Duke Breast Cancer MRI

Provider: TCIA
Collection: Duke-Breast-Cancer-MRI
URL: https://www.cancerimagingarchive.net/collection/duke-breast-cancer-mri/
DOI: https://doi.org/10.7937/TCIA.e3sv-re93
Participants: 922
Image collection size: Approximately 369 GB
License: CC BY-NC 4.0

Verified data characteristics

- Preoperative DCE-MRI.
- Radiologist lesion-location annotations.
- Clinical, demographic, pathology, treatment, and outcome spreadsheets.
- Imaging measurements and derived features.
- Selected genomic or molecular metadata.

CancerOS role

Baseline case selection, tumor localization, simulation development, and imaging pipeline evaluation.

Known issue

The collection reports that the DICOM Frame of Reference UID was replaced during de-identification and may not be reliable for alignment.

CancerOS shall not assume that all series can be registered using this UID alone.

Image orientation, position, dimensions, spacing, and registration quality must be independently checked.

DS-04 — BreastDCEDL

Provider: Zenodo / research dataset authors
URL: https://zenodo.org/records/17274053
Dataset size: 2,070 harmonized patients
License: CC BY-NC 4.0

Verified data characteristics

- Processed breast DCE-MRI from the source cohorts above.
- Standardized 3D imaging volumes.
- Segmentation masks for included subsets.
- Harmonized clinical labels.
- Defined machine-learning splits.
- NIfTI imaging data.

The processed dataset is not an independent new patient population.

The number of included subjects differs from full source collections because of data processing and selection criteria.

CancerOS role

Accelerate development of the medical imaging pipeline, particularly segmentation, preprocessing, and initial visualization.

Important temporal restriction

Different DCE phases acquired within one MRI session must not be interpreted as longitudinal patient visits.

The source study date and dynamic acquisition phase must be modeled independently.

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3. Dataset Selection Strategy

CancerOS shall maintain three distinct source pools.

Pool A — Genuine longitudinal imaging

Primary: I-SPY2.
Secondary: I-SPY1.

Purpose:

- Test actual temporal imaging navigation.
- Compare measurements across genuine visits.
- Evaluate segmentation and longitudinal measurement behavior.
- Establish limited temporal prediction baselines where adequate labels exist.

Pool B — Real baseline cases

Primary candidates: Duke and selected I-SPY cases.

Purpose:

- Provide real tumor anatomy.
- Support synthetic branching.
- Test baseline ingestion and segmentation.
- Provide clinical context for scenario eligibility.

Pool C — Processed development assets

Primary candidate: BreastDCEDL.

Purpose:

- Provide standardized NIfTI imaging.
- Supply compatible masks and metadata.
- Reduce processing requirements.
- Accelerate demonstrations.

3.1 Initial cohort targets

The intended initial selection is up to 100 distinct suitable baseline patients.

Candidate allocation:

- 40 from eligible I-SPY2 cases.

- 20 from eligible I-SPY1 cases.
- 40 from eligible Duke cases.

These are planning quotas, not established availability counts.

Actual inclusion depends on data quality, licensing, segmentation availability, and successful metadata linkage.

3.2 Exclusion rules

Exclude or quarantine cases with:

- Corrupt or unreadable imaging.
- Unresolvable patient linkage.
- Essential spatial metadata errors.
- Duplicate source identity.
- Unclear license or insufficient access permission.
- Unusable lesion annotation when required for the selected workflow.
- Incompatible or ambiguous clinical fields that cannot be safely mapped.

A patient may still be usable for a narrower task despite missing optional clinical data.

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4. Canonical Clinical Research Data Model

CancerOS shall use a normalized, patient-centered relational data model.

4.1 Core entities

Entity	Purpose
DatasetSource	Registry of research collections and licenses
SourcePatient	Pseudonymous patient mapping within a source
ResearchPatient	Internal unified research identity
CancerDiagnosis	Pathology-supported disease facts
ClinicalVisit	Actual or synthetic observation occasion

ImagingStudy	One imaging examination
ImagingSeries	One sequence or acquisition within a study
ImagingArtifact	Original or processed image file
Segmentation	Lesion mask and segmentation metadata
Lesion	Stable lesion identity where supported
TumorMeasurement	Measured or derived tumor quantities
LaboratoryResult	Dated laboratory observation
VitalSign	Dated physiological observation
SymptomObservation	Dated symptom finding
ClinicalNote	Structured or free-text physician observation
TreatmentEvent	Treatment exposure or intervention
OutcomeObservation	Recorded response or follow-up outcome
SimulationBranch	Hypothetical longitudinal descendant
SimulationRun	Reproducible simulation execution
ProvenanceRecord	Origin and transformation information
EvidenceReference	Link to clinical or research evidence
ModelResult	Derived inference or prediction
ValidationResult	Data-integrity and quality assessments

4.2 Patient identity

Required fields:

- patient_id: internal UUID.
- source_dataset_id: source registry reference.
- source_patient_key: protected, pseudonymous source identifier.
- baseline_date: source or normalized baseline timestamp.

- age_at_baseline: optional.
- sex_at_birth: optional, where provided.
- diagnosis_id: optional until verified.
- data_access_status: access control category.
- created_at.
- updated_at.

Do not store names, contact details, government identifiers, or direct patient identifiers.

Age must not be derived from an invented date of birth.

Source identifier mappings must be access-controlled and must not be exposed publicly.

4.3 Diagnosis

Fields:

- diagnosis_id
- patient_id
- diagnosis_name
- histologic_subtype
- breast_side
- grade
- clinical_stage
- pathologic_stage
- er_status
- pr_status
- her2_status
- source_reference_id
- certainty
- observed_date

Missing receptor or stage information shall be represented explicitly as unknown.

4.4 Clinical visit

Fields:

- visit_id
- patient_id
- simulation_branch_id, nullable
- visit_label
- observed_at
- relative_day
- visit_type
- provenance_type
- source_reference_id
- notes
- created_at

A clinical visit may contain several different observations acquired on different dates.

The visit date is not a substitute for each observation's actual collection timestamp.

4.5 Imaging study and series

Imaging study fields:

- study_id
- patient_id
- visit_id
- modality
- study_acquired_at
- source_study_uid
- source_dataset_id
- study_status
- provenance_type

Imaging series fields:

- series_id
- study_id
- series_uid
- sequence_name
- contrast_phase
- acquisition_time
- spacing_x_mm
- spacing_y_mm
- spacing_z_mm
- orientation
- origin
- affine_matrix
- artifact_uri
- checksum_sha256

The representation must preserve the difference between a dynamic contrast phase and a follow-up examination.

4.6 Segmentation and lesion representation

Fields:

- segmentation_id
- study_id
- lesion_id
- mask_uri
- segmentation_method
- annotation_source
- model_version

- quality_status
- spatial_reference
- provenance_type

A bounding box is not equivalent to a voxel-level segmentation mask.

The system shall label approximate segmentations accordingly.

4.7 Tumor measurements

Supported fields:

- measurement_id
- lesion_id
- study_id
- visit_id
- measurement_type
- value
- unit
- method
- measurement_uncertainty
- provenance_type
- source_reference_id

Initially supported measurements:

- Longest lesion diameter, millimeters.
- Additional orthogonal dimensions, if supported.
- Mask-derived lesion volume, cubic millimeters or cubic centimeters.
- Percentage change from baseline.
- Percentage change from prior comparable observation.

The computation must use actual voxel spacing where volumetric masks are available.

Volume is not interchangeable with longest diameter.

4.8 Laboratory results

Required fields:

- lab_result_id
- patient_id
- visit_id, optional
- collected_at
- analyte_code
- analyte_name
- value
- unit
- reference_low, optional
- reference_high, optional
- reference_source, optional
- status
- provenance_type
- source_reference_id

Initial supported analytes:

- Hemoglobin.
- White blood cell count.
- Absolute neutrophil count.
- Platelet count.
- Creatinine.
- Selected liver enzymes.
- Other approved configurable laboratory variables.

Reference ranges are context-dependent and must not be invented as universal constants.

4.9 Vital signs

Supported observations:

- Systolic and diastolic blood pressure.
- Heart rate.
- Body temperature.
- Weight.
- Other configured vital signs.

Each record shall include observation time, value, unit, and provenance.

4.10 Symptoms

Fields:

- symptom_id
- patient_id
- visit_id
- symptom_code
- severity
- severity_scale
- onset_at
- resolution_at
- provenance_type
- source_reference_id

Supported examples include fatigue, pain, nausea, and swelling.

Different symptom severity scales must not be silently combined.

4.11 Treatment events

Fields:

- treatment_event_id
- patient_id
- simulation_branch_id, optional

- category
- regimen_name, optional
- intent_or_setting, optional
- started_at
- ended_at, optional
- event_type
- recorded_status
- provenance_type
- source_reference_id

Treatment categories are exposure descriptors, not automatic evidence of response.

4.12 Outcome observations

Fields:

- outcome_id
- patient_id
- visit_id
- outcome_type
- value
- assessment_method
- assessment_date
- provenance_type
- source_reference_id

Pathologic complete response, imaging response, and overall clinical outcomes must remain distinct endpoints.

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5. File Formats and Interoperability

5.1 Source medical images

Primary accepted source:

DICOM

CancerOS shall preserve relevant spatial metadata and examination context.

Relevant elements include:

- Study and series identifiers.
- Acquisition date or approved de-identified equivalent.
- Image orientation and position.
- Pixel spacing.
- Slice geometry.
- Modality and sequence information.

5.2 Internal imaging volumes

Preferred processed format:

NIfTI (.nii.gz)

Required metadata:

- Voxel dimensions.
- Orientation.
- Image affine.
- Study identifier.
- Contrast phase.
- Provenance and transformation history.

A JSON sidecar may store processing and source metadata.

DICOM-to-NIfTI conversion must retain correct spatial geometry.

5.3 Segmentation

Supported formats:

- DICOM SEG where valid.
- NIfTI binary or labeled masks.

- Explicitly documented processed masks.

Tumor meshes may be stored as glTF/GLB or another documented geometry representation suitable for web rendering.

A generated mesh shall link to the source mask, coordinate transform, and processing version.

5.4 Clinical tabular data

Source formats:

- CSV.
- XLSX.
- JSON.
- Other explicitly supported research formats.

Canonical application representation:

- PostgreSQL relational tables.
- Typed JSON API payloads.

Parquet may be used for internal analytical batches.

5.5 Recommended units

- Time: ISO 8601 timestamps or validated date values.
- Spatial dimensions: millimeters.
- Tumor volume: cubic millimeters internally; cubic centimeters for presentation where appropriate.
- Temperature: degrees Celsius.
- Weight: kilograms.
- Blood pressure: mmHg.
- Laboratory results: measurement-specific canonical units.

Original units must remain available for audit where conversion occurs.

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6. Temporal Modeling Requirements

6.1 Multiple time axes

The system shall distinguish:

1. Actual clinical observation date.
2. Imaging acquisition date and time.
3. Treatment event date.
4. Synthetic simulation-relative day.
5. System ingestion and processing timestamp.

These values must not be substituted for one another.

6.2 Date-shifted datasets

Some research datasets may use de-identified or shifted dates.

CancerOS shall preserve valid event ordering and relative intervals while honoring source privacy requirements.

If an absolute calendar date is unreliable, the application shall prefer relative timeline labels.

6.3 Missing visits

Missing follow-up sessions are allowed.

A missing scan does not imply stable tumor disease.

A missing laboratory result does not imply a normal value.

6.4 Study comparability

Before comparing measurements, the system shall check whether the relevant images, masks, or measurement methods are comparable.

Differences in image spacing, acquisition methods, registration, and segmentation can affect measured tumor changes.

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7. Data Provenance and Lineage

Every externally sourced or generated artifact shall have a provenance record.

7.1 Provenance types

- REAL_OBSERVED
- REAL_DERIVED

- SYNTHETIC_GENERATED
- INTERPOLATED
- MODEL_PREDICTED

7.2 Mandatory provenance fields

- provenance_id
- artifact_or_record_id
- provenance_type
- source_dataset_id
- source_record_id
- parent_artifact_id, if applicable
- transformation_name
- transformation_version
- model_id, if applicable
- model_version, if applicable
- configuration_hash
- random_seed, if applicable
- created_at
- checksum_sha256
- license_reference

7.3 Lineage chain example

Real MRI study
→ normalized imaging volume
→ tumor segmentation
→ measured lesion volume
→ synthetic simulation branch
→ generated follow-up geometry
→ derived longitudinal comparison
→ AI research interpretation.

The chain must remain inspectable.

7.4 Immutable source records

Original imported data shall remain unchanged in protected storage.

Derived outputs shall be stored separately.

Corrections shall create a new version or auditable change record rather than silently replacing original information.

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8. Licensing and Data Governance

8.1 Dataset licensing

Dataset	License	Proposed handling
I-SPY2	CC BY 4.0	Permitted research development with required attribution and policy compliance
I-SPY1	CC BY 3.0	Permitted research development with required attribution and policy compliance
Duke MRI	CC BY-NC 4.0	Noncommercial use only unless additional permission is obtained
BreastDCEDL	CC BY-NC 4.0	Noncommercial use only unless additional permission is obtained

Individual components, derivative assets, and versions require separate verification.

Public availability does not eliminate licensing, privacy, or attribution obligations.

8.2 Commercial-use restriction

The hackathon prototype may use noncommercial research assets only where that use is consistent with applicable license terms.

Commercial deployment, commercialization of derived assets, sponsored use, or redistribution requires separate license review.

Any dataset whose status is uncertain shall be quarantined until its intended use is confirmed.

8.3 Source acknowledgment

The project shall maintain a source registry containing:

- Collection name.
- Collection DOI.
- Source publication.
- Dataset version.
- License.
- Retrieval date.
- Attribution requirements.
- Permitted uses.
- Redistribution restrictions.

8.4 Privacy restrictions

The project shall prohibit:

- Reidentification attempts.
- Public exposure of protected patient metadata.
- Unauthorized transmission of source images or clinical data.
- Dataset redistribution contrary to source requirements.

External LLM processing shall use the minimum necessary information and only data approved for that destination.

For the public demonstration, synthetic records or otherwise approved research data should be preferred.

8.5 Retention and deletion

Source artifacts shall remain separable from processed and synthetic outputs.

The system shall support removal of a source dataset and derived materials when required by permissions or policy.

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9. Data Processing Pipeline

The ingestion pipeline shall execute the following stages:

Stage 1 — Register source

Create dataset registry entry and verify license.

Stage 2 — Acquire approved subset

Download or import selected studies and metadata.

Stage 3 — Verify integrity

Check file completeness, format, and checksums.

Stage 4 — Resolve patient identity

Map source pseudonymous identifiers to stable internal identities.

Stage 5 — Normalize data

Standardize field names, units, dates, and coding systems.

Stage 6 — Prepare imaging

Validate DICOM geometry, convert where appropriate, and preserve source relationships.

Stage 7 — Associate tumor annotations

Link masks, bounding boxes, measurements, and derived assets.

Stage 8 — Validate clinical records

Check field types, allowed values, and source consistency.

Stage 9 — Evaluate completeness

Produce per-patient completeness and usability scores.

Stage 10 — Publish approved research case

Make the validated case available to CancerOS workflows.

Invalid records shall be quarantined and logged, not silently repaired.

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10. Training and Evaluation Partitioning

Partitioning shall occur by authentic source patient identity.

All source studies and synthetic branches belonging to a patient must remain within the same partition.

Duplicate and derivative patients across collections must be detected where source mapping permits.

The initial target split may be:

- 70% development/training patients.
- 15% validation patients.
- 15% held-out test patients.

These percentages are proposed defaults, not a mandate for small real longitudinal cohorts.

Where cohort sizes are limited, evaluation design must be adjusted transparently.

The held-out real longitudinal cohort shall not be used for simulator tuning or model selection.

Synthetic accuracy shall be reported separately from real-data accuracy.

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11. Data Quality Validation

DQ-01 — Identity integrity

No records may be linked to the wrong patient.

DQ-02 — Image integrity

Imaging volumes must retain usable orientation, spacing, dimensions, and source references.

DQ-03 — Clinical value integrity

Observed measurements shall retain valid values, units, and collection dates.

DQ-04 — Temporal integrity

Authentic clinical observations shall be ordered according to valid source timing.

DQ-05 — Provenance completeness

Every published record shall carry a recognized provenance classification.

DQ-06 — License compliance

Every dataset used shall have a verified license and approved use.

DQ-07 — Duplicate prevention

Derivative copies shall not be counted as independent patients.

DQ-08 — Missingness integrity

Missing values shall remain missing unless an explicitly identified derived or synthetic value is created.

DQ-09 — Evaluation separation

No source patient may cross the assigned training and test boundaries.

DQ-10 — Reproducibility

The team shall retain transformation versions, manifest details, and checksums necessary to recreate the curated dataset.

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12. Acceptance Criteria

The clinical data subsystem is ready when:

1. The source registry includes approved dataset entries and applicable licenses.
2. A representative sample from at least one real MRI dataset can be imported successfully.
3. Patient identity and imaging studies can be reliably linked.
4. Real and processed imaging assets retain correct spatial geometry.
5. The canonical patient and observation schemas are implemented.
6. Missing clinical variables are retained as missing.
7. A genuine patient with repeated imaging observations can be represented correctly.
8. A baseline patient can be supplied safely to the simulation engine.
9. Source artifacts and derived records have inspectable provenance.
10. Tests confirm patient-level partitioning and duplicate prevention.
11. The application can distinguish a real clinical observation from a synthetic follow-up record.
12. The dataset inventory clearly documents unavailable fields and validation limitations.

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13. Required Deliverables

The implementation shall produce:

- Source dataset inventory.
- License and attribution register.
- Canonical patient database schema.
- Clinical field dictionary.
- Imaging metadata dictionary.
- Patient and study mapping rules.
- DICOM/NIfTI ingestion pipeline.
- Data validation module.
- Provenance and lineage module.
- Research cohort manifest.
- Dataset completeness report.
- Patient-level data partition manifests.
- Representative test fixtures.
- Clinical data integration tests.

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14. Outstanding Verification Tasks

Before full data acquisition:

1. Inspect actual I-SPY2 clinical spreadsheet columns.
2. Inspect I-SPY1 clinical and measurement spreadsheets.
3. Inspect Duke clinical metadata columns and lesion annotations.
4. Confirm the exact BreastDCEDL version and compatible image files.
5. Confirm the desired use complies with each dataset's license.
6. Verify real patient visit dates and treatment metadata.
7. Confirm available blood-test and vital-sign fields.
8. Select a small cohort with usable segmentation and longitudinal measurements.
9. Establish duplicate-detection rules across source and derivative datasets.

10. Finalize the canonical code mappings for clinical variables.

These are mandatory due-diligence tasks. The dataset descriptions alone do not prove that every planned clinical variable is available.

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15. Authoritative References

1. TCIA — I-SPY2: https://www.cancerimagingarchive.net/collection/ispy2/
2. TCIA — I-SPY1: https://www.cancerimagingarchive.net/collection/ispy1/
3. TCIA — Duke Breast Cancer MRI: https://www.cancerimagingarchive.net/collection/duke-breast-cancer-mri/
4. Zenodo — BreastDCEDL: https://zenodo.org/records/17274053
5. Scientific Data — BreastDCEDL publication: https://www.nature.com/articles/s41597-026-06589-6
6. TCIA data policies: https://www.cancerimagingarchive.net/data-usage-policies-and-restrictions/

End of COS-SPEC-003 — Version 1.0